

ECO 344 - Applied Econometrics I

Chapter 5

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- 1 Introduction
- 2 Consistency
- 3 Asymptotic Normality
- 4 The Lagrange Multiplier Statistic
- 5 Asymptotic Efficiency
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Introduction

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- This chapter is all about relaxing our assumptions and making them a little more realistic.
- The price we pay is that the results only hold when the sample size is large.

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 - M.L.R - VI can be dropped completely if the sample size is large.
- M.L.R - V (Homoskedasticity, $\text{Var}(\mu|x_1, \dots, x_k) = \sigma^2$) is still needed for the usual inference. We relax it in Chapter 8.

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- In other words, the distribution of $\hat{\beta}_j$ collapses around 3 as we collect more and more data.
- The properties of OLS estimators when n approaches ∞ are called **asymptotic properties** or **large sample properties**.

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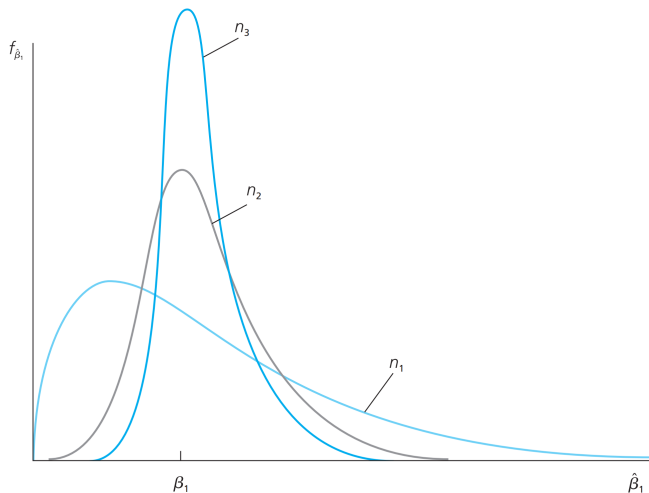
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- An estimator can be biased but still consistent.
- Consistency is the minimum we ask of an estimator. If more data does not get us closer to the truth, the estimator is of little use.

FIGURE 5.1 Sampling distributions of $\hat{\beta}_1$ for sample sizes $n_1 < n_2 < n_3$.



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- As n approaches ∞ , the distribution collapses to the single point β_1 .
- Recall from Chapter 3 that $\text{Var}(\hat{\beta}_j)$ falls as SST_j grows, and SST_j grows with n . This is why the distribution keeps shrinking.

Theorem 5.1 - Consistency of OLS

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- So the same four assumptions that give us unbiasedness also give us consistency.
- The interesting question is whether we can get consistency with **less** than M.L.R - IV.

Why is OLS Consistent?

- We know that in simple regression:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})y_i}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

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- By the law of large numbers, the numerator converges to $Cov(x, \mu)$ and the denominator converges to $Var(x)$.

$$\text{plim } \hat{\beta}_1 = \beta_1 + \frac{Cov(x, \mu)}{Var(x)}$$

M.L.R - IV' Assumption

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- This is **M.L.R - IV'** (Zero Mean and Zero Correlation).
- Now we do not need the error to be mean independent of the explanatory variables. We just need each x_j to be uncorrelated with the error.

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- It is important to note that this weaker result only applies to large samples.

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- We cannot compute this number because μ is unobserved, but we can often guess its sign.

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- The difference is that inconsistency is expressed with population variance and covariance, whereas bias uses their sample counterparts.

Asymptotic Normality

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- Note that unbiasedness and BLUE never required normality. Only exact inference did.

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- What changes with the sample size is the **sampling distribution of $\hat{\beta}_j$** , which approaches the normal distribution.
- Therefore, we can drop M.L.R - VI for large sample inference as long as the error has a finite variance.

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- The symbol $\stackrel{a}{\sim}$ reads “is approximately distributed as, in large samples”.

Theorem 5.2 - Asymptotic Normality of OLS

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- The symbol $\stackrel{a}{\sim}$ reads “is approximately distributed as, in large samples”.
- We still need homoskedasticity (M.L.R - V) here. If it fails, the usual standard errors are not valid even in large samples.

Asymptotic t-test

- As the degrees of freedom get large, the t_{n-k-1} distribution approaches the standard normal. So we can also write

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Standard Errors and the Sample Size

- Recall the estimated variance of the OLS estimator.

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- Go over Example 5.2 in the book (Standard Errors in a Birth Weight Equation).

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- Consider the model with k explanatory variables and test whether the last q of them can be excluded.

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- We only need to estimate the **restricted model** to conduct this test.

The Idea Behind the LM Test

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- So we regress $\tilde{\mu}$ on **all** the explanatory variables and look at the R-squared.
- If H_0 is true, this R-squared should be close to zero.
- The question is how large it must be before we reject. That is what the LM statistic answers.

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- 4 Under H_0 , $LM \overset{a}{\sim} \chi_q^2$. Get the critical value c from the chi-square table with q degrees of freedom. Reject the null hypothesis if $LM > c$.

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- In large samples, the LM test and the F -test rarely lead to different conclusions.
- Like the F -test, the usual LM test requires homoskedasticity (M.L.R - V).

Example 5.3 - Economic Model of Crime

- Consider the following model from Wooldridge, with $n = 2,725$.

$$\begin{aligned}narr86 = & \beta_0 + \beta_1 pcnv + \beta_2 avg\ sen + \beta_3 tottime \\ & + \beta_4 ptime86 + \beta_5 qemp86 + \mu\end{aligned}$$

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 - Regress $\tilde{\mu}$ on $pcnv$, $ptime86$, $qemp86$, $avgsen$ and $tottime$. We get $R_{\mu}^2 = 0.0015$.
 - $LM = 2,725 \times 0.0015 \approx 4.09$
 - The 10% critical value of χ^2_2 is 4.61. Since $4.09 < 4.61$, we fail to reject H_0 at the 10% level.
- A tiny R-squared can still give a sizeable LM statistic because it is multiplied by n .

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- There is a large sample version of this result.
- Under M.L.R - IV, μ is uncorrelated with any function of the explanatory variables. So there are many other consistent estimators of β_j besides OLS.
- The natural question is whether any of them is better than OLS in large samples.

Theorem 5.3 - Asymptotic Efficiency of OLS

- **Theorem:**

Under the Gauss-Markov assumptions, let $\hat{\beta}_j$ denote the OLS estimators and $\tilde{\beta}_j$ any other estimator in this class. Then for $j = 0, 1, 2, \dots, k$, the OLS estimators have the smallest asymptotic variance.

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- We say that OLS is **asymptotically efficient**.
- All the other consistent estimators in this class have an asymptotic variance at least as large.
- Homoskedasticity is essential for this result. If M.L.R - V fails, there are estimators that beat OLS (Chapter 8).

- **Consistency:** Under M.L.R - I through M.L.R - IV (or with the weaker M.L.R - IV'), $\text{plim } \hat{\beta}_j = \beta_j$.

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- **Asymptotic efficiency:** Under the Gauss-Markov assumptions, OLS has the smallest asymptotic variance.

Final Paper & Presentation

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